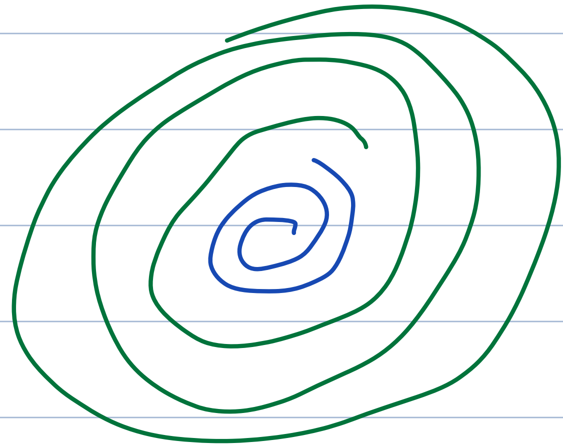
Neural network



Logistic Regression

→ straight boundary

$ax + by = c$

→ Multi-class (one vs rest)

→ Manual Feature Eng

KNN

→ Points at boundary

→ scale issue (memory)

→ Pattern learning ×

DT

→ non linear data ✓

→ thousands of feature ×

(image)

→ overfitting prone

SVM

→ non linear → kernels ✓

→ store kernel matrix (n * n)

⇓

Memory hungry

→ v slow with big data

What all we need →

NN

1. No FE
2. Handle multi-class naturally
3. scale to large datasets
4. work with high-dimention data (image, text)

Real world Application

1. FB / Google ads

→ data science → Scaler ads

→ cooking videos → zomato / swiggy etc

2) Gmail smart feature

Please find the ....

attached document

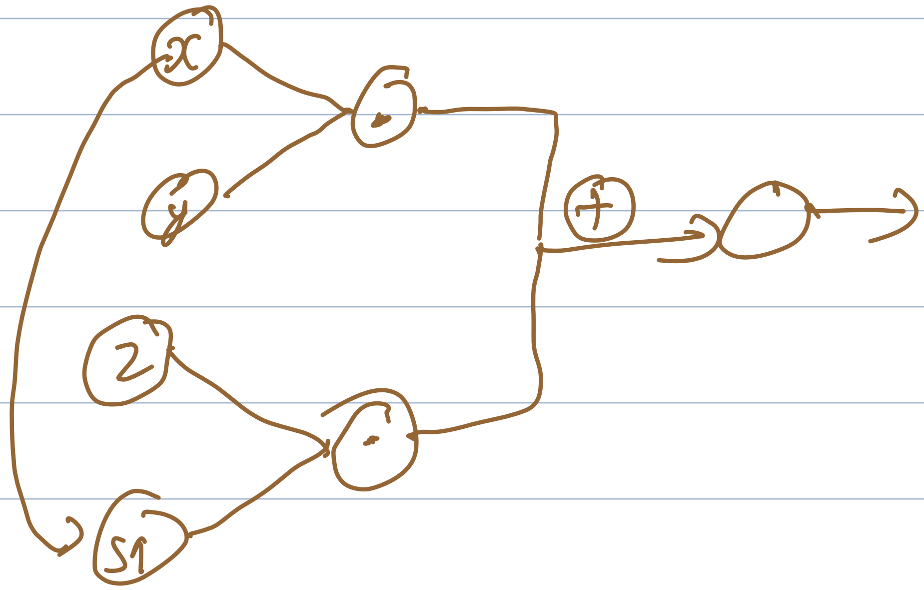
3) Photo Eraser

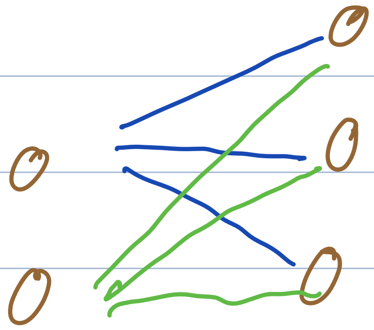
4) Data Compression

Instagram → images ↑

Auto Encoder

$$2x^2 + x \cdot y$$





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